

## APPLICATION

# Dynamic Energy Budget theory meets individual-based modelling: a generic and accessible implementation

Benjamin T. Martin<sup>1\*</sup>, Elke I. Zimmer<sup>2</sup>, Volker Grimm<sup>1</sup> and Tjalling Jager<sup>2</sup>

<sup>1</sup>Department of Ecological Modelling, Helmholtz Centre for Environmental Research – UFZ, 04318 Leipzig, Germany; and <sup>2</sup>FALW/Department of Theoretical Biology, Vrije Universiteit Amsterdam, De Boelelaan 1085, NL-1081 HV Amsterdam, the Netherlands

## Summary

1. Dynamic Energy Budget (DEB) theory was designed to understand the dynamics of biological systems from cells to populations and ecosystems via a mass balance approach of individuals. However, most work so far has focused on the level of the individual. To encourage further use of DEB theory in a population context, we developed DEB-IBM, a generic individual-based model (IBM) that is based on DEB theory.

2. The generic IBM is implemented as a computer program using NetLogo, a free software platform that is accessible to biologists with little programming background. The IBM uses DEB to represent assimilation, maintenance, growth and reproduction of individuals. The model description follows the overview, design and details (ODD) protocol, a generic format for describing IBMs, and thereby provides a novel and accessible introduction to DEB theory and how it works in a population context.

3. Dynamic Energy Budget-individual-based model can be used to explore properties of both individual life-history traits and population dynamics, which emerge from the set of DEB parameters of a species, and their interaction with environmental variables such as food density. Furthermore, DEB-IBM can be adapted to address specific research questions, for example by including spatial effects. A user manual explains how this can be done.

4. Dynamic Energy Budget-individual-based model is designed to both facilitate use and testing DEB theory in a population context and to advance individual-based modelling by basing the representation of individuals on well-tested physiological principles.

**Key-words:** bioenergetics model, Dynamic Energy Budget, individual-based model, population dynamics

## Introduction

Understanding how population dynamics emerge is one of the fundamental challenges in ecology. As the influence of individual variation, local interactions and adaptive behaviour on population dynamics has become more appreciated, individual-based models (IBMs) are playing an increasing role in both basic and applied disciplines (DeAngelis & Mooij 2005; Grimm & Railsback 2005; Stillman & Goss-Custard 2010).

IBMs represent individual organisms as unique entities that differ from each other and change over their life cycle. Individuals are characterized by a set of state variables and attributes, which are chosen according to the problem addressed with the model (Grimm *et al.* 2010). Individuals behave as autonomous entities according to behavioural rules. They interact with each other and their abiotic environment, including habitat structure and environmental drivers such as temperature, humidity or disturbances. Population dynamics emerge from these interactions.

Individual-based models have been shown to be powerful and flexible tools. However, they have also been criticized for often being based on *ad hoc* assumptions and representations of individual dynamics and behaviour (Grimm 1999). This

\*Correspondence author. E-mail: benjamin.martin@ufz.de  
DEB-IBM program and supplementary material available at: <http://cream-itn.eu/projects/wp-1/daphnia-2/deb-ibm>  
Correspondence site: <http://www.respond2articles.com/MEE/>

makes the development of IBMs inefficient and the field of individual-based modelling incoherent (Grimm & Railsback 2005). To facilitate re-usability of IBMs and their elements and to facilitate distilling general insights from specific IBMs, it is desirable to base IBMs more on standardized and well-tested approaches for individual behaviour (Berger, Hildenbrandt, & Grimm 2002).

Dynamic Energy Budget (DEB) theory (Kooijman 2010) is such an approach. It has been developed with the goal of understanding the dynamics of biological systems, from cells to ecosystems, via a balance approach for mass and energy. As in IBMs, in DEB theory, individuals are considered the key unit of interest for understanding dynamic systems at higher levels of organization. Focusing on the individual is motivated by the fact that mass and energy balances are easier to calculate for individuals than for higher or lower levels of biological complexity. Additionally, natural selection occurs at the level of the individual, which shapes the life-history traits of a species, and ultimately drives dynamics at higher levels of biological organization. DEB theory provides a quantitative framework for modelling the acquisition and use of resources for organisms over the entire life cycle. It thereby generates a quantitative explanation for the time patterns of life-history traits such as growth, maturity and reproduction in dynamic environments.

An overview of DEB theory and its applications can be found in Nisbet *et al.* (2008), Kooijman (2001), Van der Meer (2006) and Sousa, Domingos, & Kooijman (2008). A key assumption in the theory is that the mechanisms governing metabolic organization are similar among species. Therefore, the same basic model structure can be used for, in principle, all animal species; species differ in life history primarily as a result of differences in their set of DEB parameters, not because of differences in model structure. The generality of DEB facilitates a growing understanding of how life-history traits covary among and within taxa. In spite of DEB's generality, it is an empirically grounded and well-tested theory and has been applied in a range of disciplines, including ecotoxicology (Jager, Heugens, & Kooijman 2006) and aquaculture (Alunno-Bruscia, van der Veer, & Kooijman 2009), and to species from a wide range of taxonomic groups including bacteria, yeast, arthropods, fish and mammals. Yet to understand behaviour at higher levels of biological organization, tools are needed to scale from the individual model to populations.

We believe both DEB and IBMs can benefit each other; however, to date, these approaches have rarely been used in combination. Below, we discuss how each of these approaches can benefit each other and then describe the DEB-IBM framework, which we have developed to facilitate the use of DEB in an individual-based context.

### How can DEB benefit IBMs?

A common problem with the application of IBMs is their complexity. IBMs are often developed for very specific research questions, and the structure and parameterization of models defining the life history of organisms differ widely. This creates

a problem not only for model developers who often start from scratch when modelling a new species but also for the scientific community, which must try to reconcile different models or try to understand how conclusions for one species relate to another. DEB is appropriate as a building block for IBMs because it is a relatively simple model that translates environmental conditions to individual performance (growth, survival and reproduction) and is consistent with first principles such as conservation of energy. This is important because the trade-offs in life-history traits that DEB specifies (growth vs. reproduction, time and size to maturation) turn out to strongly influence population dynamics (Sæther & Bakke 2000; Denney, Jennings, & Reynolds 2002). Moreover, because DEB is a generic theory, it can be applied to virtually all species, which would facilitate broader insight from specific studies and comparisons between species.

### How can IBMs benefit DEB?

Because DEB models specify behaviour of an individual, tools are needed to extrapolate to the population level. So far, most of such population predictions based on DEB theory were made using matrix models (Klok & de Roos 1996; Klanjscek *et al.* 2006; Billoir, Péry, & Charles 2007) or the Euler–Lotka equation (Kooijman & Metz 1984; Jager *et al.* 2004). The disadvantage of these approaches is that only one state variable can be easily considered (age, stage or size), whereas the consistent application of DEB often requires considering more state variables, especially in time-varying environments. Another method for simulating population dynamics based on a model of individual performance is provided by physiologically structured population models (PSPM) [e.g. the escalator boxcar train from De Roos, Diekmann, & Metz (1992)]. PSPMs can be used to model population dynamics in dynamic environments. However, for all of these approaches (Matrix, Euler–Lotka equation, PSPM) as opposed to IBM's, variation among individuals, local interactions or adaptation cannot be easily considered in a rigorous manner. IBMs are the natural link to the population for DEB because both approaches focus on the behaviour of individuals, as a key aspect in understanding higher levels of biological complexity. Additionally, use of DEB in a population context has generally used a deterministic approach. DEB-IBM allows for the inclusion of stochasticity and thus provides a framework to investigate its effect at the population level.

### DEB-IBM links DEB theory with IBMs

Despite this potential, so far DEB theory has not been widely used in IBMs. We only know of three published examples (Kooijman, Hoeven, & Werf 1989; Alver, Alfredsen, & Olsen 2006; Bacher & Gangnery 2006). A reason for this might be that to implement DEB theory in IBMs, skills in both mathematics and computer programming are required, which many ecologists lack. Therefore, to encourage further development and use of DEB theory, we have developed a generic framework for DEB-based IBMs using a software platform that is

accessible to biologists with little programming background: NetLogo (Wilensky 1999). DEB-IBM is a generic IBM, which can be linked to specific species by using species-specific parameters. It is thus rather a framework than a specific model. We here focus on the general framework, which is designed to facilitate using DEB and IBM in combination for tackling all kinds of generic and specific questions for a wide range of species. We present a transparent and complete yet concise implementation of the DEB model for a generic isomorphic (i.e. organisms retain the same shape during growth) and ectothermic animal (Kooijman *et al.* 2008) within an IBM. In the following, we first briefly describe the DEB-IBM framework and then present the IBM and its scope.

## The DEB-IBM Framework

We implemented a scaled version of the standard DEB model as described in Kooijman *et al.* (2008). A full description of the model, following the overview, design and details (ODD) protocol for describing IBMs (Grimm *et al.* 2006, 2010), a user manual and the NetLogo file of DEB-IBM are all included in the supplementary material (<http://cream-itn.eu/projects/wp-1/daphnia-2/deb-ibm>). In the following, we provide a brief overview of DEB-IBM and describe how it can be used.

Each model individual is characterized by four primary state variables (called 'DEB state variables' hereafter) that describe the energy content of four different compartments: 'structure', which determines actual size, feeding rates and maintenance costs; 'reserves', which serve as a buffer between feeding and metabolic processes that require energy; 'maturity', a continuous state variable that regulates transitions between the three development stages (embryo, juvenile and adult) at fixed maturity levels, and a 'reproduction buffer', into which mature individuals direct energy for reproduction and which is converted into embryos during reproductive events.

In DEB theory, metabolic processes are mechanistically driven by surface/volume ratios. Individuals update their DEB state variables based on a set of differential equations. Individuals assimilate food from the environment, which enters the reserve. Energy is mobilized from the reserve and is distributed to two distinct pathways: somatic growth and maintenance on one side, and maturity maintenance, development (for immature individuals) or reproduction (for mature individuals) on the other (maintenance costs need to be satisfied first). Here,  $\kappa$  is the proportion of the mobilized energy allocated to the soma, and  $1 - \kappa$  the proportion allocated to maturity maintenance, development or reproduction. Based on the updated DEB state variables, a set of discrete events may occur. An individual dies when it cannot mobilize enough energy to pay somatic maintenance. At each time step, for each mature individual, it is calculated whether the individual has enough energy for an offspring, if it does, it produces one offspring. In the next time step of the numerical simulation, this individual is added to the population; it will start to feed exogenously when the maturity level reaches the threshold for birth; however, this default reproduction process can be easily adapted to replicate other types of reproduction behaviour. In addition to

this standard model, we have included optional submodels for the ageing process, intraspecific variation and simple predator–prey dynamics.

Species in the model are specified by the 8 'scaled' DEB parameters (see user manual), with two additional parameters for the ageing submodel (optional), and two parameters needed for the foraging submodel [you also need the two parameters ( $r$  and  $K$ ) of the logistic growth formula of the prey to run the population dynamics under logistic prey dynamic conditions].

Our implementation is compatible with a database of DEB parameters for a rapidly growing number of species: 'Add\_my\_pet' ([http://www.bio.vu.nl/thb/deb/deblab/add\\_my\\_pet/index.php](http://www.bio.vu.nl/thb/deb/deblab/add_my_pet/index.php)). In the user manual, we provide a detailed explanation of how to obtain parameters from this database and input them into DEB-IBM. The 'Add\_my\_pet' database is relatively new, with parameters values for approximately 60 species, with varying degree of support. However, users can assess the degree of support for a species in the database, because the data used to derive the parameter set for each species are given in a corresponding data file, and within the file the references from which the data were taken are listed.

Many users still will have to obtain DEB parameters themselves. There are currently two thorough reviews and guides for parameterizing a DEB model for a species (Van der Meer 2006; Kooijman *et al.* 2008). If data are very limiting, a general set of parameters can be estimated from maximum body size of an individual (Kooijman *et al.* 2008). While users can cope with less, generally data for growth and reproduction at multiple food densities provides enough information to get a good set of parameters for use in DEB-IBM. Parameterization tools, DEBtool (available in both Matlab and the free software Octave) and DEBtoxM (specific for toxic stress, Matlab only), can be obtained from <http://www.bio.vu.nl/thb/deb/deblab/>, which perform the required optimization techniques for varying levels of data availability. This level of use requires deeper investment into DEB theory.

There are two levels of application for our generic framework. First, it can be used to explore properties of both individual life-history traits and population dynamics, which emerge from the set of DEB parameters of a species, and their interaction with environmental variables such as food density. For this, no programming or technical understanding of DEB theory are required. Users need only input DEB parameters and environmental conditions in the graphical user interface, from which they can monitor and record various individual and population-level output such as fecundity, population density and size structure.

The second level of use, involves adapting DEB-IBM to address a specific research question. For this, users must learn how to change the code of the generic model. For example, the research question might be: how are the population dynamics of a species influenced by changes in land use? In this case, the user would adapt the generic DEB-IBM to include space and movement behaviour of individuals, with DEB theory acting as the energetic model for the individual. We provide detailed examples of how the model can be adapted to include both

spatial and behavioural aspects. The NetLogo implementation is flexible enough to add all kinds of modules or alter existing ones, including ones that are in DEBtool or DEBtoxM.

This more advanced use of the model requires users to learn programming in NetLogo. However, NetLogo is an exceptionally well-documented software platform that was specifically designed for implementing IBMs; moreover, a recent textbook on individual-based modelling, which is based on NetLogo is available (Railsback & Grimm 2012). NetLogo comes with powerful built-in procedures, leading to a shallow learning curve. This makes both IBMs and DEB more accessible to ecologists without formal training in computer programming. NetLogo has some limitations, in particular regarding computation time, number of agents and spatial units it can deal with, and the lack of a tool for debugging the software. However, these limitations turn out to not be constricting for many models and problems in population ecology. Moreover, because NetLogo slowly but surely is turning into a standard platform for implementing IBMs, we expect that these limitations will be overcome in the near future (see, for example, the recent link between NetLogo and the computationally more powerful platform RePast: [http://repast.sourceforge.net/repast\\_simulation.html](http://repast.sourceforge.net/repast_simulation.html)).

For DEB-IBM, we did not choose a general programming language such as C++ or Java because learning these languages to the point where users can implement or modify IBMs would usually be too time-consuming for most ecologists. Likewise, we considered none of the alternative software platforms, e.g. Repast ([repast.sourceforge.net](http://repast.sourceforge.net)) or MASON (Luke *et al.* 2005) suitable because they are much harder to learn and not as thoroughly documented (for comparative reviews of software platform for individual-based or agent-based models, see Railsback, Lytinen, & Jackson 2006; Nikolai & Madey 2009).

## Discussion

Dynamic Energy Budget-IBM can be used without modification to make general estimations of population characteristics, such as population growth rate, in simple environments, and as a learning tool for understanding how the physiological properties of individuals can influence population dynamics. While other tools, such as matrix models or the Euler–Lotka equation can be used to estimate population growth rates in constant environments, they cannot as easily be extended to dynamic environments. A further advantage of DEB-IBM is that we can consider the interactions between a predator population and its prey. In the default version of DEB-IBM, we have allowed the option to model dynamic predator–prey population dynamics, assuming the prey follows a logistic growth pattern and is depleted via predation; however, this can be adapted when needed to model prey dynamics of a specific system. Thus, DEB-IBM can be used to estimate carrying capacity of the predator population as a function of its environment. Because DEB-IBM predicts dynamics in time, it lends itself to more rigorous testing with population-level data, which often consist of time series observations of population

density and/or size structure. This is important because validation of models against population data is necessary to build confidence in the model for applied uses.

It should be noted that although DEB-IBM facilitates applying DEB theory in individual-based population models, using it still requires commitment. For specific research questions, DEB-IBM merely serves as a starting point. Researchers will have to consider species-specific processes such as the rules for converting the reproduction buffer into offspring. In the generic model, individuals reproduce when they have enough energy to produce one offspring. However, many animals produce clutches of offspring, either at fixed time intervals or when triggered by an environmental cue. These differences in life-history strategy can easily be incorporated into the generic model, and in the user manual, we give examples of how to do so. Additionally, relevant behaviours such as dispersal or habitat selection may have to be considered. Users of DEB-IBM should thus be prepared to learn basic skills in NetLogo, but this requires, owing to the design and excellent documentation of NetLogo, usually not more than a few days.

Like any useful theory, DEB theory is not static, and there are still plenty of open questions within DEB that require dedicated research. A growing international community is currently working with this theory, so we can expect new developments in the near future. One benefit of using DEB in a population context is that it highlights aspects of the individual dynamics, which are especially relevant for population dynamics. Often these are areas that have been overlooked by those focusing solely on individuals. Our own initial use of DEB-IBM has highlighted important questions where further research is needed. For example, within DEB, a general pattern of intraspecific variation in parameter values has been suggested (Kooijman, Hoeven, & Werf 1989); however, little research to date has investigated how DEB parameters (co)vary among individuals within a population. Additionally, little research has so far been carried out on the process of starvation. Kooijman (2010) offers some possibilities to handle starvation within a DEB context, but these rules are probably highly species-specific and require further evaluation.

Nevertheless, the advantage of using a mechanistic framework like DEB is that once these questions are addressed, and the major processes understood, they are more likely to apply in untested conditions, whereas phenomenological approaches can only be applied within the range of tested conditions. Additionally, research on starvation within a DEB context may help shed light on how similar the mechanisms of the starvation process are among a wide range of taxa. DEB has a lot to offer for solving specific problems, but to exploit its benefits as a general theory, it needs to be used and tested more widely at the population level. This would increase confidence in the model, clarify its limitations and possibly lead to further improvement.

## Acknowledgements

This research has been financially supported by the European Union under the 7th Framework Programme (project acronym CREAM, contract number

PITN-GA-2009-238148.) We would like to thank Sandrine Charles and three anonymous reviewers for their extremely thorough review of this article and the entire supplementary material and for many insightful suggestions and comments.

## References

- Alunno-Bruscia, M., van der Veer, H.W. & Kooijman, S.A.L.M. (2009) The AquaDEB project (phase I): analysing the physiological flexibility of aquatic species and connecting physiological diversity to ecological and evolutionary processes by using Dynamic Energy Budgets. *Journal of Sea Research*, **62**, 43–48.
- Alver, M., Alfredsen, J. & Olsen, Y. (2006) An individual-based population model for rotifer (*Brachionus plicatilis*) cultures. *Hydrobiologia*, **560**, 93–108.
- Bacher, C. & Gangnery, A. (2006) Use of dynamic energy budget and individual based models to simulate the dynamics of cultivated oyster populations. *Journal of Sea Research*, **56**, 140–155.
- Berger, U., Hildenbrandt, H. & Grimm, V. (2002) Towards a standard for the individual-based modeling of plant populations: self-thinning and the field-of-neighborhood approach. *Natural Resource Modelling*, **15**, 39–54.
- Billoir, E., Péry, A.R.R. & Charles, S. (2007) Integrating the lethal and sub-lethal effects of toxic compounds into the population dynamics of *Daphnia magna*: a combination of the DEBtox and matrix population models. *Ecological Modelling*, **203**, 204–214.
- De Roos, A.M., Diekmann, O. & Metz, J.A.J. (1992) Studying the dynamics of structured population models: a versatile technique and its application to *daphnia*. *The American Naturalist*, **139**, 123–147.
- DeAngelis, D.L. & Mooij, W.M. (2005) Individual-based modelling of ecological and evolutionary processes. *Annual Review Ecology, Evolution, and Systematics*, **36**, 147–168.
- Denney, N.H., Jennings, S. & Reynolds, J.D. (2002) Life-history correlates of maximum population growth rates in marine fishes. *Proceedings of the Royal Society B*, **269**, 2229–2237.
- Grimm, V. (1999) Ten years of individual-based modelling in ecology: what have we learned and what could we learn in the future?. *Ecological Modelling*, **155**, 129–148.
- Grimm, V. & Railsback, S.F. (2005) *Individual-based Modeling and Ecology*. Princeton University Press, Princeton, NJ.
- Grimm, V., Berger, U., Bastiansen, F., Eliassen, S., Ginot, V., Giske, J., Goss-Custard, J., Grand, T., Heinz, S., Huse, G., Huth, A., Jepsen, J.U., Jørgensen, C., Mooij, W.M., Müller, B., Pe'er, G., Piu, C., Railsback, S.F., Robbins, A.M., Robbins, M.M., Rossmanith, E., Rüger, N., Strand, E., Souissi, S., Stillman, R.A., Vabø, R., Visser, U. & DeAngelis, D.L. (2006) A standard protocol for describing individual-based and agent-based models. *Ecological Modelling*, **198**, 115–126.
- Grimm, V., Berger, U., DeAngelis, D.L., Polhill, G., Giske, J. & Railsback, S.F. (2010) The ODD protocol: a review and first update. *Ecological Modelling*, **221**, 2760–2768.
- Jager, T., Heugens, E.H.W. & Kooijman, S.A.L.M. (2006) Making sense of ecotoxicological test results: towards application of process-based models. *Ecotoxicology*, **15**, 305–314.
- Jager, T., Crommentuijn, T., van Gestel, C.A.M. & Kooijman, S.A.L.M. (2004) Simultaneous modeling of multiple end points in life-cycle toxicity tests. *Environmental Science & Technology*, **38**, 2894–2900.
- Klanjscek, T., Caswell, H., Neubert, M.G. & Nisbet, R.M. (2006) Integrating Dynamic Energy Budgets into matrix population models. *Ecological Modelling*, **196**, 407–420.
- Klok, C. & de Roos, A.M. (1996) Population level consequences of toxicological influences on individual growth and reproduction in *Lumbricus rubellus* (Lumbricidae, Oligochaeta). *Ecotoxicology and Environmental Safety*, **33**, 118–127.
- Kooijman, S.A.L.M. (2001) Quantitative aspects of metabolic organization; a discussion of concepts. *Philosophical Transactions of the Royal Society B*, **356**, 331–349.
- Kooijman, S.A.L.M. (2010) *Dynamic Energy Budget theory for Metabolic Organisation*. Cambridge University Press, Cambridge.
- Kooijman, S.A.L.M., Hoeven, N.V.D. & Werf, D.C.V. (1989) Population consequences of a physiological model for individuals. *Functional Ecology*, **3**, 325–336.
- Kooijman, S.A.L.M. & Metz, J.A.J. (1984) On the dynamics of chemically stressed populations: the deduction of population consequences from effects on individuals. *Ecotoxicology and Environmental Safety*, **8**, 254–274.
- Kooijman, S.A.L.M., Sousa, T., Pecquerie, L., Van der Meer, J. & Jager, T. (2008) From food-dependent statistics to metabolic parameters, a practical guide to the use of Dynamic Energy Budget theory. *Biological Reviews*, **83**, 533–552.
- Luke, S., Cioffi-Revilla, C., Panait, L., Sullivan, K. & Balan, G. (2005) MASON: a multiagent simulation environment. *Simulation*, **81**, 517–527.
- Nikolai, C. & Madey, G. (2009) Tools of the trade: a survey of various agent based modeling platforms. *Journal of Artificial Societies and Social Simulation*, **12**, 2. <http://jasss.soc.surrey.ac.uk/12/2/2.html>.
- Nisbet, R.M., Muller, E.B., Lika, K. & Kooijman, S.A.L.M. (2008) From molecules to ecosystems through Dynamic Energy Budget models. *Journal of Animal Ecology*, **69**, 913–926.
- Railsback, S.F. & Grimm, V. (2012) *Agent-based and Individual-based Modeling: A Practical Introduction*. Princeton University Press, Princeton, N.J.
- Railsback, S.F., Lytinen, S.L. & Jackson, S.K. (2006) Agent-based simulation platforms: review and development recommendations. *Simulation*, **82**, 609–623.
- Sæther, B.E. & Bakke, Ø. (2000) Avian life history variation and contribution of demographic traits to the population growth rate. *Ecology*, **81**, 642–653.
- Sousa, T., Domingos, T. & Kooijman, S.A.L.M. (2008) From empirical patterns to theory: a formal metabolic theory of life. *Philosophical Transactions of the Royal Society B*, **363**, 2453–2464.
- Stillman, R.A. & Goss-Custard, J.D. (2010) Individual-based ecology of coastal birds. *Biological Reviews*, **85**, 413–434.
- Van der Meer, J. (2006) An introduction to Dynamic Energy Budget (DEB) models with special emphasis on parameter estimation. *Journal of Sea Research*, **56**, 85–102.
- Wilensky, U. (1999) *NetLogo*. <http://ccl.northwestern.edu/netlogo/>. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

Received 1 June 2011; accepted 7 November 2011

Handling Editor: Robert Freckleton